

GSC-X Nuclear Sandbox

Discovery of Nuclear Operating Corridors Through Adaptive Executive Optimization

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<https://gds-gsc-x.com>

GitHub Repository

<https://github.com/sisacek66198068/GSC-X>

Related GSC-X Sandbox Portfolio

GSC-X Industrial Sandbox

GSC-X Reliability Sandbox

GSC-X Fusion Sandbox

GSC-X Spaceflight Sandbox

GSC-X Battery Sandbox

GSC-X Robotics Sandbox

GSC-X Nuclear Sandbox

GSC-X Energy Network Sandbox

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Abstract

This report presents the first comprehensive validation campaign of the GSC-X Nuclear Sandbox.

The study evaluates adaptive executive optimization across synthetic nuclear-transmutation digital twins representing multiple operating principles, including baseline operation, accelerator-driven behavior, fast high-burn behavior, molten-salt-like smoothing, and several GSC-X hybrid executive regimes.

The primary contribution of this study is not the identification of a single optimal operating point.

The primary contribution is the discovery of a structured Nuclear Operating Map containing multiple viable operating corridors with distinct trade-offs between transmutation performance, safety, and energy efficiency.

Across TRUE N10000 validation campaigns, the GSC-X hybrid regimes revealed safe-performance, industrial-performance, balanced-adaptive, and eco-efficient corridors.

The strongest safety transformation was observed in the transition from the high-burn FAST regime to the **FAST_HYBRID** regime, where high transmutation performance was preserved while unsafe operation was eliminated and energy demand was substantially reduced.

The strongest energy-efficiency result was observed in the **GSCX_MSR_HYBRID** corridor, which achieved the best yield-per-energy while maintaining zero unsafe time.

This work should be interpreted as a synthetic digital-twin control validation and operating-corridor discovery study, not as a claim of real nuclear waste elimination or real nuclear-physics validation.

1 Introduction

Advanced nuclear waste-processing and transmutation concepts face several competing objectives.

These include maximizing transmutation throughput, maintaining operational safety, reducing energy demand, and preserving long-term stability under uncertain operating conditions.

A common engineering intuition is that stronger burn intensity should produce better waste-processing performance.

The Nuclear Sandbox investigates a different hypothesis.

Rather than searching only for maximum burn intensity, this study asks whether adaptive executive optimization can discover previously hidden operating corridors inside nuclear-transmutation digital twins.

The objective is not to propose a new reactor technology.

The objective is not to claim direct nuclear waste elimination.

The objective is to evaluate whether the GSC-X architecture can function as a complementary executive optimization layer above future digital twins, advanced transmutation concepts, and nuclear-processing simulations.

In this framing, GSC-X does not replace nuclear physics, reactor engineering, or transport simulation.

Instead, it attempts to discover and maintain operating regimes that balance performance, safety, and energy efficiency.

2 GSC-X Nuclear Executive Architecture

The nuclear branch of GSC-X consists of:

- Context Memory (CM)
- Early Limit Indicator (ELI)
- Adaptive flux scheduling
- Executive pulse regulation
- Hybrid operating-mode stabilization
- Safety-corridor monitoring
- Operating-corridor discovery

CM provides temporal awareness of accumulated system state and operating history.

ELI estimates proximity to future instability boundaries before unsafe operation emerges.

Together these layers form an adaptive executive optimization framework designed to identify and maintain useful operating corridors rather than merely increasing raw transmutation intensity.

The architecture is therefore best understood as an executive discovery layer.

Its purpose is to search the operating landscape and identify regimes where throughput, energy demand, and safety form favorable trade-offs.

3 Validation Portfolio

The Nuclear Validation campaign includes three main validation stages.

3.1 Digital Twin V1.6

Digital Twin V1.6 compared a baseline synthetic transmutation schedule against the GSCX_CM-ELI executive mode.

This stage produced the first positive ECO-oriented result.

The benchmark was validated on TRUE N10000 runs.

3.2 Digital Twin V2.1 – ECO Corridor Discovery

Digital Twin V2.1 expanded the operating portfolio to include multiple synthetic nuclear-processing archetypes:

- BASE
- ADS
- FAST
- MSR
- GSCX_CM-ELI
- GSCX_MSR-HYBRID

This stage confirmed the emergence of an ECO-efficient corridor through the GSCX_MSR-HYBRID mode.

3.3 Digital Twin V3 – Nuclear Operating Map

Digital Twin V3 extended the benchmark into a complete Nuclear Operating Map.

The tested modes were:

- BASE
- ADS
- ADS-HYBRID
- FAST
- FAST-HYBRID
- MSR
- GSCX_CM-ELI
- GSCX_MSR-HYBRID

This validation stage demonstrated that the operating space contains several distinct corridors rather than one universal optimum.

4 Discovery of Nuclear Operating Corridors

The key result of the Nuclear Sandbox is the emergence of multiple operating corridors rather than a single best regime.

Corridor	Representative Mode	Strategic Objective
Maximum Burn	FAST	Highest raw transmutation
Safe Performance	FAST_HYBRID	High transmutation with zero unsafe time
Industrial Performance	ADS_HYBRID	Strong safe processing performance
Balanced Adaptive	GSCX_CM_ELI	Active adaptive control compromise
ECO Efficient	GSCX_MSR_HYBRID	Highest yield per unit energy

Table 1: Nuclear operating corridors identified during V3 TRUE N10000 validation.

This result changes the interpretation of the Nuclear Sandbox.

The benchmark does not simply compare a classical controller against GSC-X.

Instead, it maps a synthetic nuclear-processing landscape across different operating philosophies.

The central finding is that stronger burn is not always the best operational objective.

Several useful regimes exist, each optimized for a different strategic purpose.

5 Digital Twin V1.6 TRUE N10000

Digital Twin V1.6 provided the first robust evidence that an adaptive GSC-X executive layer could improve both toxicity reduction and yield-per-energy within the synthetic transmutation model.

Metric	BASE	GSCX_CM_ELI
Toxicity Reduction	0.223560	0.226727
Remaining Fraction	0.776440	0.773273
Converted Mass Proxy	0.957117	0.971171
Energy	65.498465	66.354658
Yield per Energy	0.014613	0.014637
Unsafe Time	0.000000	0.000000

Table 2: Digital Twin V1.6 TRUE N10000 comparison between BASE and GSCX_CM_ELI.

5.1 Interpretation

The V1.6 result was modest but important.

The GSCX_CM_ELI mode improved toxicity reduction, converted mass proxy, and yield-per-energy while preserving zero unsafe time.

This stage established the first stable positive direction for the Nuclear Sandbox.

6 Digital Twin V2.1 TRUE N10000

Digital Twin V2.1 introduced a broader set of operating archetypes.

The objective was to determine whether the synthetic transmutation landscape contains distinct operating regimes rather than a single controller comparison.

Mode	Toxicity Reduction	Converted Mass	Energy	Yield/Energy	Unsafe
BASE	0.231046	1.159050	72.531580	0.015994	0.000000
ADS	0.365570	1.829621	181.947249	0.010062	0.000500
FAST	0.396214	1.983859	329.096469	0.006035	116.587900
MSR	0.241601	1.211746	68.943482	0.017587	0.000000
GSCX_CM_ELI	0.259580	1.303045	78.697921	0.016586	0.000000
GSCX_MSR_HYBRID	0.235892	1.183809	60.554386	0.019563	0.000000

Table 3: Digital Twin V2.1 TRUE N10000 operating archetype comparison.

6.1 ECO Corridor

The most important V2.1 result was the emergence of the GSCX_MSR_HYBRID ECO corridor. This was the first validation stage that revealed a distinct operational corridor rather than a simple controller comparison.

Metric	BASE	GSCX_MSR_HYBRID
Toxicity Reduction	0.231046	0.235892
Converted Mass Proxy	1.159050	1.183809
Energy	72.531580	60.554386
Yield per Energy	0.015994	0.019563
Unsafe Time	0.000000	0.000000

Table 4: ECO corridor comparison between BASE and GSCX_MSR_HYBRID.

Compared with BASE, the GSCX_MSR_HYBRID mode achieved:

- higher toxicity reduction
- higher converted mass proxy
- lower energy demand
- higher yield-per-energy
- zero unsafe time

This result suggests that adaptive executive scheduling may be particularly valuable for discovering energy-efficient transmutation corridors.

7 Digital Twin V3 TRUE N10000

Digital Twin V3 expanded the benchmark into a complete Nuclear Operating Map.

This stage introduced hybridized ADS and FAST regimes in addition to the earlier BASE, MSR, and GSC-X modes.

Mode	Toxicity Reduction	Converted Mass	Energy	Yield/Energy	Unsafe
BASE	0.231141	1.160099	72.578423	0.015999	0.000000
ADS	0.365451	1.829293	181.770621	0.010070	0.000300
ADS_HYBRID	0.323956	1.622754	113.962008	0.014246	0.000000
FAST	0.396142	1.982585	329.130705	0.006030	116.664400
FAST_HYBRID	0.341591	1.710142	116.923946	0.014634	0.000000
MSR	0.241591	1.212487	68.950704	0.017596	0.000000
GSCX_CM_ELI	0.259757	1.303119	78.788218	0.016568	0.000000
GSCX_MSR_HYBRID	0.236769	1.187961	60.817936	0.019547	0.000000

Table 5: Nuclear Operating Map V3 TRUE N10000.

8 Safe Performance Corridor

The strongest safety transformation was observed in the transition from FAST to FAST_HYBRID.

FAST produced the highest raw transmutation performance, but it was energy-intensive and unsafe.

The FAST_HYBRID mode preserved high transmutation performance while eliminating unsafe operation and greatly reducing energy demand.

Metric	FAST	FAST_HYBRID
Toxicity Reduction	0.396142	0.341591
Converted Mass Proxy	1.982585	1.710142
Energy	329.130705	116.923946
Yield per Energy	0.006030	0.014634
Unsafe Time	116.664400	0.000000

Table 6: Transformation of the high-burn FAST regime into the safe-performance FAST_HYBRID corridor.

8.1 Interpretation

This is one of the strongest results of the Nuclear Sandbox.

The transformation from FAST to FAST_HYBRID suggests that GSC-X does not merely push the system harder.

Instead, it can reorganize an unsafe high-burn regime into a practical safe-performance corridor.

The result preserves a large fraction of transmutation performance while eliminating unsafe behavior. This observation motivated the transition from controller benchmarking toward operating-corridor mapping in later Nuclear Sandbox development.

9 Industrial Performance Corridor

The ADS_HYBRID mode formed an industrial-performance corridor.

Compared with ADS, it reduced energy demand and eliminated unsafe time while preserving strong transmutation performance.

Metric	ADS	ADS_HYBRID
Toxicity Reduction	0.365451	0.323956
Converted Mass Proxy	1.829293	1.622754
Energy	181.770621	113.962008
Yield per Energy	0.010070	0.014246
Unsafe Time	0.000300	0.000000

Table 7: ADS compared with the safer and more energy-balanced ADS_HYBRID operating corridor.

10 Balanced Adaptive Corridor

The GSCX_CM_ELI mode represents a balanced adaptive corridor.

It does not maximize raw burn.

It does not minimize energy demand.

Instead, it provides an active compromise with improved transmutation performance, improved yield-per-energy relative to BASE, and zero unsafe time.

Metric	BASE	GSCX_CM_ELI
Toxicity Reduction	0.231141	0.259757
Converted Mass Proxy	1.160099	1.303119
Energy	72.578423	78.788218
Yield per Energy	0.015999	0.016568
Unsafe Time	0.000000	0.000000

Table 8: Balanced adaptive corridor represented by GSCX_CM_ELI.

11 ECO Efficient Corridor

The GSCX_MSR_HYBRID mode represents the ECO-efficient corridor.

It achieved the highest yield-per-energy in the V3 TRUE N10000 operating map while maintaining zero unsafe time.

Metric	BASE	GSCX_MSR_HYBRID
Toxicity Reduction	0.231141	0.236769
Converted Mass Proxy	1.160099	1.187961
Energy	72.578423	60.817936
Yield per Energy	0.015999	0.019547
Unsafe Time	0.000000	0.000000

Table 9: ECO-efficient corridor represented by GSCX_MSR_HYBRID.

11.1 Interpretation

Rather than maximizing transmutation intensity, the ECO corridor maximizes useful transmutation per unit energy.

This corridor may be especially relevant for future digital-twin optimization studies where efficiency, safety, and long-term operational stability are more important than maximum burn alone.

12 Figures

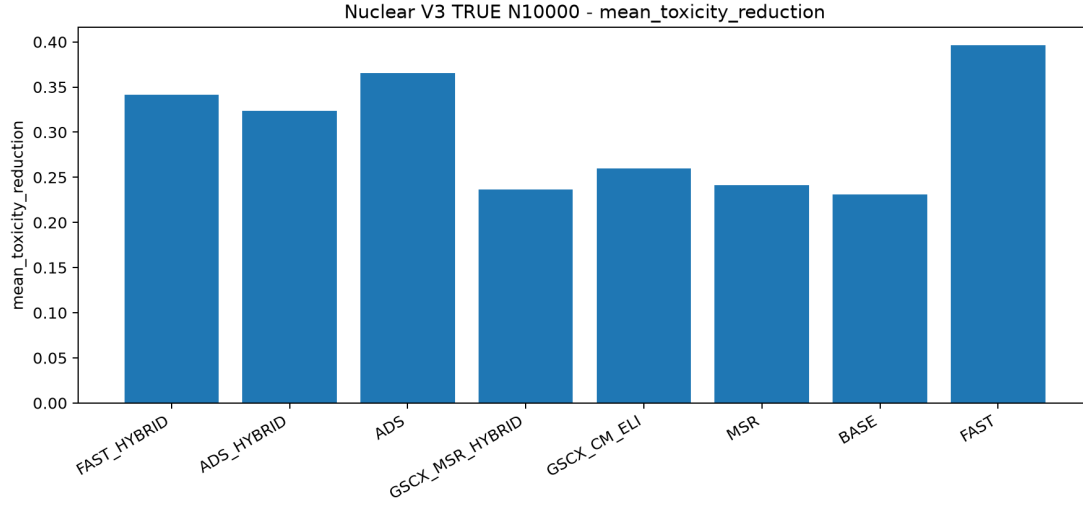


Figure 1: Comparison of mean toxicity reduction across the Nuclear Operating Map V3 TRUE N10000 validation. FAST achieved the highest raw transmutation performance, while FAST_HYBRID and ADS_HYBRID preserved strong performance under safe operating conditions.

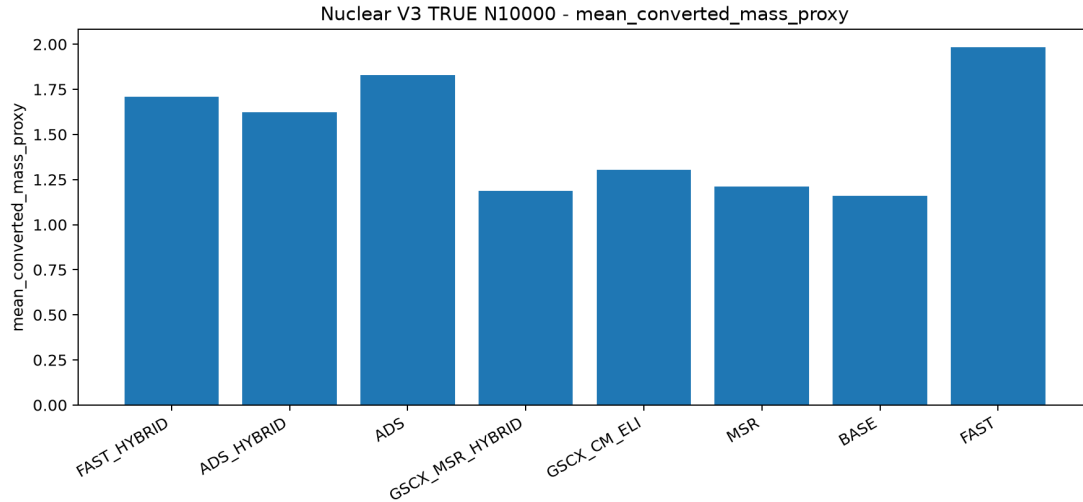


Figure 2: Comparison of mean converted mass proxy across the Nuclear Operating Map V3 TRUE N10000 validation. FAST delivered the highest conversion throughput, while hybrid operating corridors retained substantial processing capability with improved efficiency and safety.

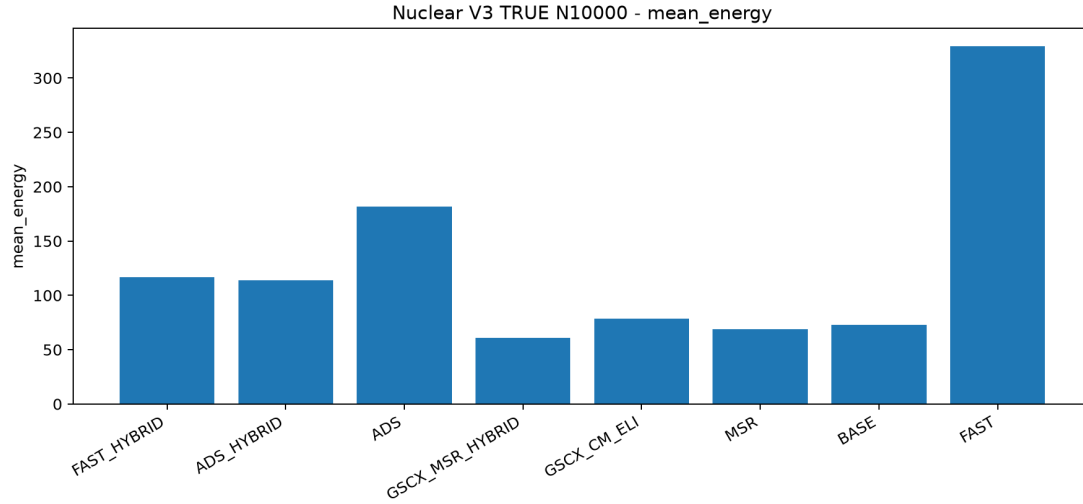


Figure 3: Comparison of mean energy demand across the Nuclear Operating Map V3 TRUE N10000 validation. Hybrid executive regimes significantly reduced energy requirements relative to aggressive high-burn operating modes.

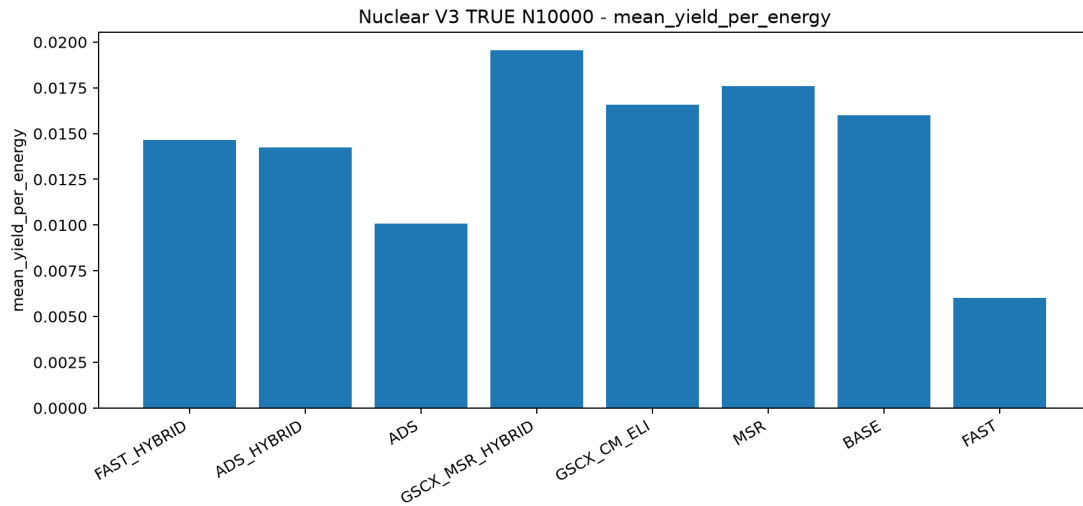


Figure 4: Comparison of mean yield-per-energy across the Nuclear Operating Map V3 TRUE N10000 validation. GSCX_MSR_HYBRID achieved the highest energy efficiency while maintaining zero unsafe time.

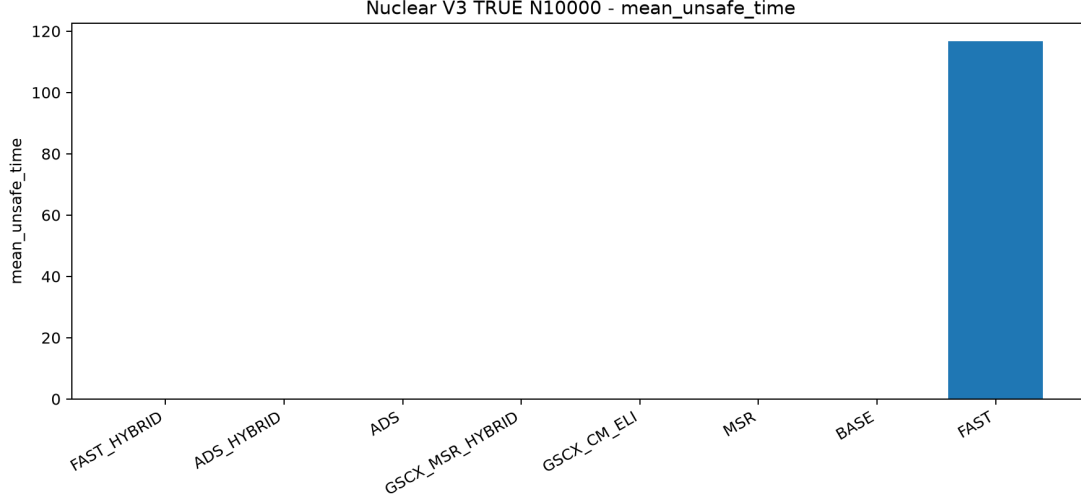


Figure 5: Comparison of mean unsafe operating time across the Nuclear Operating Map V3 TRUE N10000 validation. FAST exhibited substantial unsafe exposure, whereas all hybrid executive corridors maintained effectively zero unsafe time.

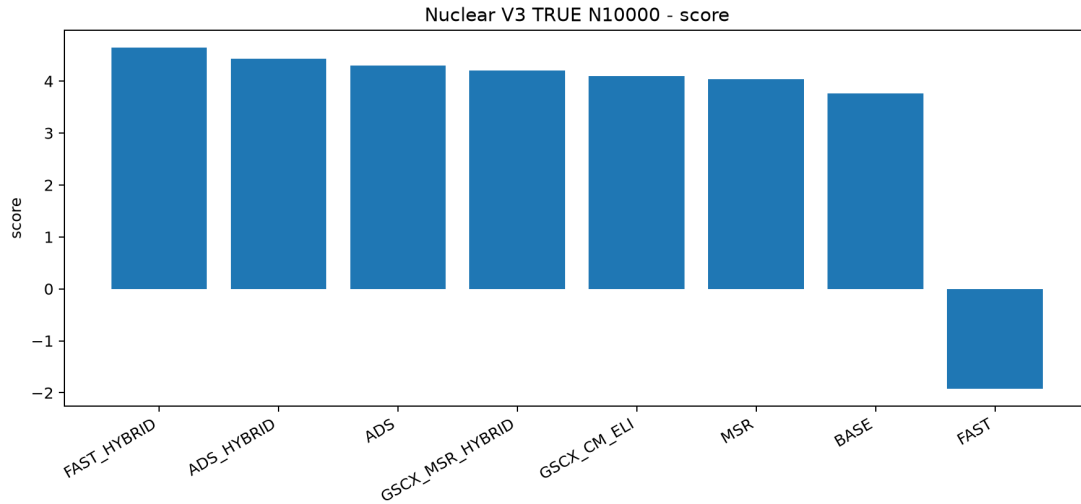


Figure 6: Composite performance score across the Nuclear Operating Map V3 TRUE N10000 validation. The highest-ranking operating corridors emerged from FAST_HYBRID and ADS_HYBRID, reflecting favorable trade-offs between performance, energy demand, and operational safety.

13 Discussion

The Nuclear Sandbox suggests that the most important result is not a single winning controller.

The most important result is the discovery of a structured operating landscape.

FAST remains the maximum-burn regime, but it is unsafe and energy-heavy.

FAST_HYBRID creates a safe-performance corridor by preserving high transmutation performance while eliminating unsafe behavior.

ADS_HYBRID creates a safe industrial-performance corridor.

GSCX_CM_ELI provides a balanced adaptive corridor.

GSCX_MSR_HYBRID provides an ECO-efficient corridor with the highest yield-per-energy and

lower energy demand than BASE. The principal outcome of the Nuclear Sandbox is therefore not the identification of a single superior operating regime, but the discovery of a structured Nuclear Operating Map containing multiple corridors optimized for different strategic objectives.

These findings suggest that future nuclear-processing digital twins may benefit not only from stronger physical concepts, but from adaptive executive layers capable of discovering and maintaining useful operating corridors. The Nuclear Operating Map may be interpreted as an executive optimization landscape rather than a single-controller benchmark.

14 Potential Operational Implications and Future Economic Impact

The validation results presented in this study quantify technical metrics including toxicity reduction, converted mass proxy, energy demand, yield-per-energy, and unsafe time.

Although direct economic modelling was outside the scope of the present work, the observed emergence of safe-performance and ECO-efficient corridors suggests potentially meaningful operational implications.

Area of Impact	Potential GSC-X Mechanism	Observed Technical Effect	Potential Operational Implication
Waste Processing Efficiency	Operating-corridor discovery and adaptive executive scheduling	Improved yield-per-energy in GSCX-MSR_HYBRID	Potential improvement in energy-normalized waste-processing performance
Safety-Oriented Operation	ELI-based avoidance of unsafe operating regimes	Unsafe time reduced to zero in FAST_HYBRID	Potential reduction of unstable high-burn operating exposure
High-Throughput Processing	Hybridization of aggressive burn regimes	FAST_HYBRID preserved high transmutation performance with lower energy	Potential safe-performance corridor for future digital-twin studies
Industrial Deployment Pathway	Executive optimization above existing concepts	ADS_HYBRID formed a safe industrial-performance corridor	Potential support for future transmutation-system optimization studies
Digital-Twin Optimization	GSC-X as a platform-independent executive layer	Multiple operating corridors discovered across TRUE N10000 validation	Potential use as an optimization layer above future nuclear digital twins

Table 10: Potential operational implications inferred from validated synthetic digital-twin results. Real economic impacts require domain-specific assessment.

These observations should be interpreted as engineering implications derived from validated synthetic digital-twin metrics rather than direct economic or physical nuclear measurements.

Future studies may quantify domain-specific impacts using OpenMC, ENDF/B data, operational records, isotope-specific reaction networks, and reactor-specific digital twins.

15 Limitations

This study is a synthetic digital-twin control validation.

It does not include:

- OpenMC transport calculations

- ENDF/B nuclear data libraries
- real neutron spectra
- real isotope reaction networks
- real reactor geometry
- real accelerator-driven system physics
- real molten-salt reactor physics

Therefore, the results should not be interpreted as proof of real nuclear waste elimination.

They should be interpreted as evidence that adaptive executive optimization can discover useful operating corridors inside synthetic nuclear-transmutation digital twins.

16 Conclusion

The GSC-X Nuclear Sandbox demonstrates that adaptive executive optimization can reveal a structured Nuclear Operating Map across synthetic transmutation environments.

The central contribution of this work is the discovery of Nuclear Operating Corridors, demonstrating that multiple viable optimization targets may coexist within the same synthetic transmutation environment.

Rather than identifying a single optimum, the V3 TRUE N10000 validation revealed a family of operating corridors spanning maximum-burn, safe-performance, industrial-performance, balanced-adaptive, and eco-efficient regimes.

The strongest safety transformation was achieved by FAST_HYBRID, which transformed an unsafe high-burn regime into a safe-performance corridor.

The strongest efficiency result was achieved by GSCX_MSR_HYBRID, which produced the highest yield-per-energy while maintaining zero unsafe time.

These results support continued investigation of GSC-X as a complementary executive optimization layer for future nuclear digital twins, transmutation simulations, and advanced waste-processing studies.

17 Future Work

- OpenMC prototype integration
- ENDF/B-supported isotope behavior
- realistic isotope-chain modelling
- neutron-spectrum-aware optimization
- accelerator-driven system digital twins
- molten-salt transmutation digital twins
- fast-spectrum transmutation scenarios
- hardware-in-the-loop and operator-assistance studies
- collaboration with nuclear research institutions

An important future objective is to determine whether the operating-corridor structure observed in the synthetic Nuclear Sandbox persists under increasingly realistic nuclear-physics simulations.

The GSC-X architecture is intended to operate as a complementary predictive executive layer above existing or future digital-twin and simulation environments rather than replacing established nuclear-engineering platforms.

Acknowledgements

This research utilized ChatGPT and Gemini as analytical and computational assistants during development, validation, analysis, and documentation of the presented results.

Reproducibility and Public Repository

Validation artifacts, benchmark summaries, logs, figures, and public project documentation are available at:

<https://github.com/sisacek66198068/GSC-X>

Only validation artifacts are published. Source code is not included in the public validation package due to patent-pending and intellectual-property protection considerations.

Interactive Demonstrator

An interactive Nuclear Sandbox presenting the validation results, benchmark portfolio, and digital-twin demonstrations described in this study is available at:

<https://nuclear.gds-gsc-x.com>

References

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